

Client:

```
GNU nano 8.7.1 client.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>

#define FIFO1 "client_to_server"
#define FIFO2 "server_to_client"

int main()
{
    char message[100];
    int fd1, fd2;

    fd1 = open(FIFO1, O_WRONLY);
    strcpy(message, "Hello Server!");
    write(fd1, message, strlen(message) + 1);
    close(fd1);

    fd2 = open(FIFO2, O_RDONLY);
    read(fd2, message, sizeof(message));
    printf("Server: %s\n", message);
    close(fd2);

    return 0;
}
```

server

```
GNU nano 8.7.1 server.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>

#define FIFO1 "client_to_server"
#define FIFO2 "server_to_client"

int main()
{
    char message[100];
    int fd1, fd2;

    mkfifo(FIFO1, 0666);
    mkfifo(FIFO2, 0666);

    printf("Server waiting for client...\n");

    fd1 = open(FIFO1, O_RDONLY);
    read(fd1, message, sizeof(message));
    printf("Client: %s\n", message);
    close(fd1);

    fd2 = open(FIFO2, O_WRONLY);
    strcpy(message, "Hello Client, message received!");
    write(fd2, message, strlen(message) + 1);
    close(fd2);

    unlink(FIFO1);
    unlink(FIFO2);

    return 0;
}
```

Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ nano client.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ gcc client.c -o client
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ nano server.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ gcc server.c -o server
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ ./server
Server waiting for client...
Client: Hello Server!
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ nano client.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ |
```

client

```

uday@DESKTOP-47L1GTF:~$ ls
2520090089_OSSP Before_fork.save OSLAB OsLab status
uday@DESKTOP-47L1GTF:~$ cd 2520090089_OSSP
uday@DESKTOP-47L1GTF:~/2520090089_OSSP$ ls
2520090089_Practical 2520090089_Skill 'File Permission (1).pdf' ForgeOS permission permission.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP$ cd 2520090089_Practical
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ ls
Practical-01 Practical-03 Practical-05 Practical-07 Practical-09 Signals
Practical-02 Practical-04 Practical-06 Practical-08 Practical-10 'signals ossp (1).pdf'
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ cd practical-06
-bash: cd: practical-06: No such file or directory
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ cd Practical-06
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ ./client
Server: Hello Client, message received!
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ █

```

Signal:

```

GNU nano 8.7.1 signal.c
#include <stdio.h>
#include <stdlib.h>
#include <signal.h>
#include <unistd.h>

void handle_signal(int sig)
{
    if (sig == SIGINT)
        printf("\nSIGINT received!\n");

    else if (sig == SIGTERM)
        printf("SIGTERM received!\n");

    else if (sig == SIGUSR1)
        printf("SIGUSR1 received!\n");
}

int main()
{
    signal(SIGINT, handle_signal);
    signal(SIGTERM, handle_signal);
    signal(SIGUSR1, handle_signal);

    printf("Process ID: %d\n", getpid());
    printf("Waiting for signals...\n");

    while (1)
    {
        sleep(1);
    }

    return 0;
}

[ Read 33 lines ]

```

```

uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ SIGUSR1 084
SIGUSR1: command not found
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ kill - SIGUSR1 084
-bash: kill: : invalid signal specification
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ KILL -SIGUSR1 1084
KILL: command not found
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ kill -SIGUSR1 1084
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ kill -SIGTERM 1084
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ kill -9 1084
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ █

```

Output:

```

uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ nano server.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ nano signal.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ gcc signal.c -o signal
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ ./signal
Process ID: 1084
Waiting for signals...
SIGUSR1 received!
SIGTERM received!
Killed
./signal
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ nano signal.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-06$ █

```